E. G. S. PILLAY ENGINEERING COLLEGE

(Autonomous)

Approved by AICTE, New Delhi| Affiliated to Anna University, Chennai Accredited by NAAC with A++ Grade | Accredited by NBA (CSE, ECE, IT)

NAGAPATTINAM – 611002

B.E. Biomedical Engineering

Second Year– Third Semester

Course Code	Course Name	L	T	P	C	Maximum Marks		
						CA	ES	Total
Theory Course								
1901MA301	Engineering Mathematics - III (Transforms And Linear Algebra)	3	2	0	4	40	60	100
1902CS305	Object Oriented Programming and Data Structures	3	0	0	3	40	60	100
1902BM301	Fundamentals of Biochemistry	3	0	0	3	40	60	100
1902BM302	Biomedical Circuits and Networks	3	2	0	4	40	60	100
1902BM303	Bio Sensors and Measurements	3	0	0	3	40	60	100
1902BM304	Human Anatomy and Physiology	3	0	0	3	40	60	100
Laboratory Course								
1902BM351	Devices and Circuits Laboratory	0	0	2	1	50	50	100
1902BM352	Biochemistry and Human Physiology Laboratory	0	0	2	1	50	50	100
1902CS353	C++ and Data Structures Laboratory	0	0	2	1	50	50	100
1904GE351	Life Skills :Soft Skill	0	0	2	1	100	-	100
Mandatory Course								
1901MCX02	Constitution of India	2	0	0	0	0	0	0
	Total	**20**	**4**	**8**	**24**	**490**	**510**	**1000**

L–Lecture| T–Tutorial| P–Practical| C–Credit| CA–Continuous Assessment |ES–End Semester

1901MA301	TRANSFORMS AND LINEAR ALGEBRA ENGINEERING MATHEMATICS - III	L	T	P	C
		3	1	0	4

PREREQUISITE:

1. Basic knowledge in Differentiation
2. Basic knowledge in Integration

COURSEOBJECTIVES:

1. To introduce solving systems of linear equations, Matrix operations.
2. To familiarize Vector spaces and subspaces; linear independence and span of a set of vectors, basis and dimension; the standard bases for common vector spaces.
3. To introduce the effective mathematical tools for the solutions of partial differential equations that model several Physical processes and to develop Z transform techniques for discrete time systems.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	-	-	-	-	-	-	-	2
CO2	3	3	3	-	-	-	-	-	-	-	-	2
CO3	2	2	2	-	-	-	-	-	-	-	-	2
CO4	3	3	3	-	-	-	-	-	-	-	-	2
CO5	3	3	3	-	-	-	-	-	-	-	-	2
AVG	2.8	2.6	2.6	-	-	-	-	-	-	-	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	1	-
CO2	1	2	-
CO3	2	1	-
CO4	2	1	-
CO5	3	2	-
AVG	1.8	1.5	-

MODULEI — VECTOR SPACES — 12 Hours

Vector spaces–Subspaces–Linear combinations and system of Linear equations–Linear independence and Linear dependence – Bases and Dimensions

MODULE II — LINEAR TRANSFORMATIONS — 12 Hours

Linear combination system of linear equation–algebra of transformation–Linear transformation of matrices–Linear functional – transpose of linear transformation

MODULE III — FOURIER SERIES — 12 Hours

Dirichlet’s conditions–General Fourier series–Odd and even functions–Half range sine-series–Half range cosine series– Parseval’s identity – Harmonic analysis.

MODULE IV — FOURIER TRANSFORMS — 12 Hours

Statement of Fourier integral theorem–Fourier transform pair–Fourier sine and cosine transforms–Properties–Transforms of simple functions – Convolution theorem – Parseval‟s identity

MODULEV — Z–TRANSFORMS AND DIFFERENCE EQUATIONS — 12 Hours

Z-transforms–Elementary properties–Inverse Z–transform using partial fraction and residues–Convolution theorem–
Formation of difference equations–Solution of difference equations using Z–transform.

TOTAL 60 HOURS

FURTHER READING/CONTENT BEYOND SYLLABUS / SEMINAR:

1. Numerical Solution of non-homogeneous partial differential equations

COURSE OUTCOMES:

After completion of the course, Student will be able to

CO1	Understand the concepts of vector spaces, subspaces, bases, and dimension.=
CO2	Relate matrices and linear transformations, compute Eigen values and Eigen vectors of linear transformations.
CO3	Use Fourier series analysis, which is central to many applications in engineering.
CO4	Apply Fourier transform techniques in a wide variety of situations.
CO5	Apply Z-transform techniques for discrete-time systems.

REFERENCES:

1. Friedberg, A.H., Insel ,A.J. and Spence,L.,―Linear Algebra‖, Prentice-Hall of India,NewDelhi,2004.
2. Veerarajan. T.,“ Transforms and Partial Differential Equations”, Second reprint, Tata McGraw Hill Education Pvt. Ltd., New Delhi, 2012
3. Kumaresan,S.,―Linear Algebra–A geo metric approach‖, Prentice–Hall of India, New Delhi,Reprint,2010.
4. Grewal. B.S., “Higher Engineering Mathematics”,42nd Edition, Khanna Publishers,Delhi,2012.
5. Bali.N.P and Manish Goyal,“A Textbook of Engineering Mathematics”,7th Edition, Laxmi Publications Pvt Ltd,2007
6. Ramana. B.V.“Higher Engineering Mathematics”, Tata Mc- Graw Hill Publishing Company Limited, New Delhi,2008.
7. Narayanan.S.Manica vachagom Pillay. T.K and Ramanaiah.G“ Advanced Mathematics for Engineering Students”Vol. II & III, S.Viswanathan Publishers Pvt Ltd. 1998.
8. www.nptelvideos.in/2012/11/mathematics-iii.html

1902CS305	OBJECT ORIENTED PROGRAMMING AND DATA STRUCTURES (Common to B.E/B.Tech -All branches)	L	T	P	C
		3	0	0	3

Course Objectives:

1. To comprehend the fundamentals of object oriented programming, particularly in C++.
2. To use object oriented programming to implement data structures.
3. To introduce linear, non-linear data structures and their applications.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	-	-	-	-	-	-	-	-	2
CO2	3	3	1	-	-	-	-	-	-	-	-	2
CO3	3	3	1	-	-	-	-	-	-	-	-	2
CO4	3	3	1	-	-	-	-	-	-	-	-	-
CO5	3	3	1	-	-	-	-	-	-	-	-	-
AVG	3	3	1	-	-	-	-	-	-	-	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	3	-
CO2	-	3	-
CO3	-	-	-
CO4	-	-	-
CO5	-	-	-
AVG	-	3	-

Unit I	OBJECT ORIENTED PROGRAMMING	9 Hours

Evolution of Programming methodologies-Introduction to OOP-Basic features-Structure of C++Program- Compiling and Executing C++ Program - Data types - Operators - Expressions - Control statements &Iteration statements in C++ -Arrays-Structures-Pointers

Unit II	FUNCTIONS & CONSTRUCTORS	9 Hours

Functions-Passing Data to Functions-Scope and Visibility of variables in Functions-Dynamic Binding-data members - member functions -this Pointer – Friend Functions –Friend Classes - Constructors and Destructors.

Unit III	LINEAR DATA STRUCTURES	9 Hours

Abstract Data Types(ADTs)–List ADT–array-based implementation–linked list implementation—single linked lists –Polynomial Manipulation - Stack ADT – Queue ADT - Evaluating arithmetic expressions

Unit IV	NON-LINEAR DATA STRUCTURES	9 Hours

Trees–Binary Tree-Binary search trees-Tree traversal-Expression manipulation-Symbol table construction-AVL trees: Rotation, Insertion, Deletion,–Red black tree – Graph and its representations – Graph Traversals –Representation of Graphs–Breadth-first search–Depth-first search-Connected components.

Unit V	SORTING and SEARCHING	9 Hours

Sorting Techniques-Selection, Bubble, Insertion, Merge, Heap, Quick, and Radixsort - Address calculation-Linear search-Binary search-Hash table methods.

	Total:	45 Hours

Further Reading:	
	JAVA Program
	Advanced Sorting Algorithms.
Course Outcomes:	
	After completion of the course, Student will be able to
	CO1: Understand various programming methodologies and Object-Oriented Programming (OOP) concepts.
	CO2: Understand the scope and application of functions in real-time problems.
	CO3: Design algorithms to solve real-life problems using data structures.
	CO4: Recognize the usage of non-linear data structures such as Binary Search Trees, AVL trees, and Heap trees in applications.
	CO5: Analyze various sorting and searching algorithms.

References:

1. Deitel and Deitel,—C++,How To Program‖, Seventh Edition, Pearson Education,2013.
2. Mark Allen Weiss, —Data Structures and Algorithm Analysis in C++‖, Fourth Edition, Addison-Wesley, 2013.
3. Bhushan Trivedi, —Programming with ANSIC++,A Step-By-Step approach ‖,Oxford University Press, 2010.
4. Goodrich, MichaelT., Rober toTamassia, David Mount,—Data Structures and Algorithms in C++‖,7th Edition, Wiley. 2016.
5. Thomas H.Cormen, Charles E.Leiserson, Ronald L. Rivest and Clifford Stein, " Introduction to Algorithms", Third Edition, Mc Graw Hill, 2009.
6. Bjarne Strou strup,—The C++Programming Language‖, 3rd Edition, Pearson Education,2007.
7. Ellis Horowitz, Sartaj Sahni and Dinesh Mehta,—Fundamentals of Data Structures in C++‖,Galgotia Publications, 2007.

1902BM301	FUNDAMENTALS OF BIOCHEMISTRY	L	T	P	C
		3	2	0	3

COURSE OBJECTIVES:

1. To understand about the functioning of physiological system.
2. To analyse the biochemical reactions and the various methods.
3. To know the significance of bio molecules in biological systems.
4. To give an introduction about the clinical diseases
5. To understand the concepts of Enzymes and Kinetics.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	1	-	-	-	-	-	-	2
CO2	3	2	1	1	1	-	-	-	-	-	-	2
CO3	3	2	1	1	1	-	-	-	-	-	-	-
CO4	3	2	1	1	1	-	-	-	-	-	-	-
CO5	3	2	1	1	1	-	-	-	-	-	-	-
AVG	3	2	1	1	1	-	-	-	-	-	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	3	2
CO2	-	3	2
CO3	-	-	-
CO4	-	-	-
CO5	-	3	2
AVG	-	3	2

UNIT I	INTRODUCTIONTO BIOCHEMISTRY	9 Hours

Bio molecules, structure of water & its importance–Important non covalent forces–Hydrogen bonds, electrostatic, hydrophobic & vander waals forces–Acid, base &buffers–pH, Henderson Hassel Balch equation. Biological buffers and Their significance–Principle of viscosity–surface tension, adsorption, diffusion, osmosis & their applications in biological systems.

UNIT II	BIO-ENERGETICS	9 Hours

High energy compounds-electro negative potential of compounds, respiratory chain-ATP cycle, Calculation of ATP during oxidation of glucose and fatty acids. DNA-RNA, Proteins, Lipids, Carbohydrates.

UNITIII	MACRO OLECULES, VITAMINS, HORMONES, ENZYMES	9 Hours

Physical and chemical properties-structure of hemoglobin: immune globulin and nucleo protein, classification and their properties, occurrence, functions, requirements, deficiency manifestations and role of vitamins as coenzyme, chemical Nature and properties, hormones, Nomenclature-enzyme kinetics-classification and their properties, mechanism of action, enzyme induction and inhibition, coenzyme significance and enzymes of clinical importance.

UNITIV	CLINICAL DISEASES	9 Hours

Diabetes mellitus- insulin dependent diabetes mellitus, non-insulin dependent diabetes mellitus, measurement of HbA1clevels-atherosclerosis, fatty liver, and obesity-hormonal disorders, aging, inborn errors of metabolism organ Function tests		
UNITV	**ENZYME AND ITS KINETICS**	**9 Hours**
Classification of enzymes: apo enzyme, coenzyme, holo enzyme and cofactors. Kinetics of enzymes - Michaelis, Menten equation. Factors affecting enzymatic activity: temperature, pH, substrate concentration and enzyme concentration. Inhibitors of enzyme action-Competitive, non-competitive, irreversible. Enzyme-Mode of action, all osteric and covalent regulation. Clinical significance of enzymes. Measurement of enzyme activity and interpretation of units		
	Total:	**45 Hours**

FURTHER READING:

- Study metabolic pathways in pathological conditions.
- Assess the significance of bio molecules in biological systems.
- Analyze the etiology and biological parameters in metabolic diseases.

COURSE OUTCOMES:

After completion of the course, Student will be able to

CO1:learn about the structure and function of biomolecules and how they interact within biological systems

CO2:Explain how energy is produced and used in the body, analyze metabolic pathways, and apply bioenergetics principles to biological systems

CO3:Understand and compare the Physical and chemical properties and structure of hemoglobin, immunoglobulin and nuclei protein.

CO4:Compare the pathological conditions like obesity, Diabetes mellitus, atherosclerosis, fatty liver, and hormonal disorders, aging.

CO5:Analyze the clinical significance of enzyme activity and kinetics.

TEXTBOOKS:

1. Lehninger A.L. ,Nelson D.L. and Cox M.M. Principles of Biochemistry. CB Spublishers and distributers, 2010

2. Thomas M.Devlin. Textbook of Biochemistry with clinical correlations. Wiley Liss Publishers David L.Nelson, Michael M. Cox, Lehninger ―Principles of Biochemistry Macmillan, 6th edition 2013.

REFERENCES:

1. CO5: Analyze various sorting and searching algorithms. Murray R.K.,Granner D.K., Mayes P.A.and Rod well V.W.Harpers Biochemistry. Applet on and Lange, Stanford, Conneticut,2012
2. Keith Wilson and John Walker, ―Practical Biochemistry – Principles & Techniques Oxford University press, 7th Edition, 2010.
3. Trevorpalmer - Understanding Enzymes ‖,Ellis HorwoodLTD,4rd Edition,1995.

1902BM302	BIOMEDICAL CIRCUITS AND NETWORKS	L	T	P	C
		3	0	0	4

COURSE OBJECTIVES:

1. Ability in identifying passive and active circuit elements/components and basic knowledge on their operation and application.
2. To prepare the students to have a basic knowledge in the analysis of Electric Networks
3. To understand and solve the given circuit with various theorems and methods and to distinguish between tie set and cut set methods for solving various circuits.
4. To gain knowledge about coupled circuits.
5. In depth knowledge in Integral & Differential Calculus and fundamental knowledge on Laplace Theorem & its inverse.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C204.1	3	3	2	2	-	-	-	-	-	-	-	1
C204.2	3	3	2	2	-	-	-	-	-	-	-	1
C204.3	3	3	2	2	-	-	-	-	-	-	-	1
C204.4	3	3	2	2	-	-	-	-	-	-	-	-
C204.5	3	3	2	2	-	-	-	-	-	-	-	-
AVG	3	3	2	2	-	-	-	-	-	-	-	1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
C204.1	-	1	1
C204.2	-	1	1
C204.3	-	1	1
C204.4	-	1	-
C204.5	-	-	-
AVG	-	1	1

UNITI — MESH CURRENT AND NETWORK ANALYSIS — 9Hours

Kirchhoff’s Voltage Law- Formulation of Mesh Equations- Solution of mesh equations by Cramer’s rule and matrix method, Driving point impedance, Transfer impedance.

UNIT II — NODALAN ALYSIS OF CIRCUITS — 9 Hours

Graph of Network: Concept of Tree Branch, Tree link, junctions, Incident matrix, Tie-set matrix, Cutset matrix, determination of loop current and node voltages.

UNIT III — RESONANT CIRCUITS — 9 Hours

Series and Parallel Resonance, Impedance and Admittance Characteristics -Quality Factor, Half-Power Points, Bandwidth, Resonant voltage rise, Transform diagrams

UNIT IV — NETWORK ANALYSIS — 9 Hours

Kirchoff’s Current Law- Formulation of node equations and solutions- Driving point admittance, Transfer admittance, Solutions of Problems with DC and AC sources. Definition and implications of Superposition Theorem, Thevenin’s Theorem- Norton’s Theorem- Reciprocity Theorem- Compensation Theorem- Maximum Power Transfer Theorem- Millman’s Theorem, Star-Delta transformations, Solutions and Problems with DC and AC sources.

UNIT V — COUPLED CIRCUITS — 9 Hours

Magnetic Coupling-polarity of coils, polarity of induced voltage, concept of self and mutual inductance, coefficient of coupling, Solution of Problems Circuit Transients-DC Transient in R-L&R-C circuits with and Without initial charge, R-L-C circuits, AC transients in sinusoidal RL,R-C,&R-L-C circuits.

	Total:	**45 Hours**

FURTHERREADING:

- Understand, Describe and Analyze the Transients in electrical networks and solve related problems.
- Apply Laplace Transform and form Transfer Function for different kinds of electrical networks for analyzing them and solve related problems

COURSEOUTCOMES:

After completion of the course, Students will be able to

CO1:Calculate the circuit parameters using mesh and nodal method
CO2:Apply various network theorems for the analysis of electrical circuits.
CO3:Examine the concept of resonance and coupled circuits
CO4:Determine transient response of DC circuits.
CO5:Construct the graphs of networks

TEXTBOOKS:

1. Electric Circuits and Networks - a text book written by Suresh Kumar K S.,Published by Pearson Education ISBN:9788131713907 Pages: 840
2. Circuit Theory and Network by S.p.Ghosh and A.K.Chakraborty, 2011

REFERENCES:

1. Neamen Donald A., Electronics Ckt. Analyzer & Design, Tata Mc Graw Hill.
2. Boylestad Robert L. ,Nashelsky Louis, Electronics Devices & Circuits, Pearson Education.

1902BM303	BIOSENSORS AND MEASUREMENTS	L	T	P	C
		3	0	0	3

COURSE OBJECTIVES:

1. To Understand the Units and Standards of measurements for various physical quantities and how to use the measurement for calibration and error analysis.
2. To analyze the Characteristics of the Transducer using their models and Responses.
3. To make an experiment on various Resistance type transducer using their principle of operation and Applications.
4. To gain knowledge about Bio Sensors and their Applications.
5. To acquire the knowledge of another special Transducer.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	-	-	-	-	-	-	-	2
CO2	3	2	1	-	-	-	-	-	-	-	-	2
CO3	3	2	1	-	-	-	-	-	-	-	-	2
CO4	3	2	1	-	-	-	-	-	-	-	-	2
CO5	3	2	1	-	-	-	-	-	-	-	-	2
AVG	3	2	2	-	-	-	-	-	-	-	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	-	3
CO2	-	-	-
CO3	-	-	-
CO4	-	-	-
CO5	-	-	3
AVG	-	-	3

UNIT I — SCIENCE OF MEASUREMENT — 9 Hours

Units and Standards-calibration methods, statics calibration. classification of errors-error analysis, statistical methods -odds and un certainity

UNIT II — CHARACTERISTICS OF TRANSDUCERS — 9 Hours

Static characteristics-accuracy, precision, sensitivity, linearity. mathematical model of transducers–zero, first order and second - order transducers - response to impulse step, ramp and sinusoidal inputs

UNIT III — VARIABLERESISTANCETRANSDUCERS — 9 Hours

Resistance Potentiometer –Principle of operation, construction details, characteristics and applications-strain gauges- resistance thermometers- thermistors- hot-wire anemometer and humidity sensors.

UNIT IV — BIOSENSORS-PHYSIOLOGICAL RECEPTORS — 9 Hours

Type of Bio Sensor-Chemo receptors, Baro receptors, Touch receptors, Biosensors-Working Principle and Applications

UNIT V — SPECIAL TRANSDUCERS — 9 Hours

Piezo electric transducers , magneto strictive transducer, IC sensor digital transducers-smart sensor-fibre optic transducers-Introduction to MEMS and Nano Sensors

	Total:	45Hours

FURTHER READING:

- Bio receptors and Bio detectors
- DNA Sequencing with nano pores

COURSE OUTCOMES:

After completion of the course, Student will be able to

1.Explain the Science of Measurement and Error Analysis
2.Identify the characteristics of transducers and its Responses
3.Experiment with Variable Resistance Transducers and their Applications.
4.Describe the working function of Different types of Bio Sensors and their applications
5.Explain the working principles of special Transducers.

TEXTBOOKS:

1. L.A Geddes and L.E.Baker Principles of Applied Biomedical Instrumentation‟‟ Third Edition,– John Wiley and sons, Reprint 2008.
2. Albert D.Helfrick and William D.Cooper.―Modern Electronic Instrumentation and Measurement Techniques Prentice Hall of India, 2007.

REFERENCES:

1. S.M.Sze, - Semiconductor Sensors,‖NewYork,1994, John Wiley & Sons.
2. L. Ristic, - Sensor Technology and Devices,‖1994,Artech House, Inc.
3. John G. Webster,HalitEren―Measurement, Instrumentation, and Sensors Handbook: Electromagnetic, Optical, Radiation, Chemical, and Biomedical Measurement‖, 2017
4. Jacob Fraden,―Handbook of Modern Sensors: Physics, Designs, and Applications‖, Fourth Edition, Springer

1902BM304	HUMANAN ATOMY AND PHYSIOLOGY	L	T	P	C
		3	0	0	3

COURSE OBJECTIVES:

1. Know basic structural and functional elements of human body.
2. Learn organs and structures involving in system formation and functions.
3. Understand all systems in the human body.
4. Gain knowledge about sensory system
5. Better understanding of fluid maintenance

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	-	-	-	3	-	-	-	2
CO2	3	2	1	-	-	-	-	3	-	-	-	2
CO3	3	2	1	-	-	-	-	-	-	-	-	-
CO4	3	2	1	-	-	-	-	-	-	-	-	-
CO5	3	2	1	-	-	-	-	-	-	-	-	-
AVG	3	2	2	-	-	-	-	3	-	-	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	-	-
CO2	-	-	-
CO3	-	-	-
CO4	-	-	-
CO5	-	-	-
AVG	-	-	-

UNIT I | **BASIC ELEMENTS OF HUMAN BODY** | **9 Hours**

Cell-Structure and organelles, Functions of each component in the cell. Cell membrane–transport across membrane, Origin of cell membrane potential. Tissue-Types, Specialized tissues, functions.

UNIT II | **RESPIRATORY SYSTEM AND URINARY SYSTEM** | **9 Hours**

Respiratory System- Components of respiratory system , Respiratory Mechanism, Types of respiration, Oxygen and carbon dioxide transport and acid base regulation. Urinary system: Structure of Kidney and Nephron. Mechanism of Urine formation, Urinary reflex, Homeo stasis and blood pressure regulation by urinary system. Digestive system: Structure of Digestive system-Parts of Digestive system-Digestive process.

UNIT III | **BLOOD AND CARDIO VASCULAR SYSTEM** | **9 Hours**

Blood composition - functions of blood, functions of RBC, WBC types and their functions Blood groups, importance of blood groups, identification of blood groups. Blood vessels-Structure of heart–Properties of Cardiacmuscle, Conducting System of heart, Cardiac cycle, Heart sound, Volume and pressure changes and regulation of heart rate, Coronary Circulation. Factors regulating Blood flow.

UNIT IV | **SKELETAL AND SPECIAL SENSORY SYSTEM** | **9 Hours**

Skeletal system: Bone types and functions, Axial Skeleton and Appendicular Skeleton. Joint- Types of Joint, Cartilage structure, types and functions. Special Sensory system- Eye, Ear and Skin - diseases and related surgery.

UNIT V | **NERVOUS SYSTEM** | **9 Hours**

Structure of a Neuron – Types of Neuron. Neuroglial Cells, Synapses and types. Brain – Divisions of brain lobes, Cross Sectional Anatomy of Brain, Cortical localizations and functions. Spinal cord – Tracts of spinal cord, Spinal Nerve, Reflex mechanism– Types of reflex, Autonomic nervous system and its functions.

Total:	**45 Hours**

FURTHER READING:

- 1.To determine hemoglobin count in the blood by Sahlis method.
- 2.In-vitro recognition of A,B,O blood groups by slide test.
- 3.To find the total Red Blood Cell count using Neubauer shaemo cytometer.
- 4.To find the total White Blood Cell count using Neubauer shaemo cytometer.
- 5.TostudyECG Machine

COURSE OUTCOMES:
After completion of the course, Student will be able to

CO1:	Describe the basic structural and functional elements of the human body.
CO2:	Explain the components and mechanisms of the respiratory and urinary systems.
CO3:	Identify organs and structures involved in system formation and their functions.
CO4:	Recognize the importance and functions of the skeletal system and special sensory systems.
CO5:	Elucidate the structure of neurons, their types, and the anatomy of the brain.

TEXTBOOKS:

1. Eldra Pearl Solomon."Introduction to Human Anatomy and Physiology", W.B.Saunders Company,2003.
2. Frank H.Netter, Atlas of human anatomy ‖, Netter basic science,7th edition 2019.

REFERENCES:

1. William F. Ganong ," Review of Medical Physiology ‖,Mc Graw Hill, New Delhi,25th Edition,2015.
2. Arthur C.Guyton," Text book of Medical Physiology", Elsevier Saunders,11th Edition,2006
3. Elaine.N. Marieb, ―Essential of Human Anatomy and Physiology‖, Pearson Education New Delhi, 8th Edition, 2007.
4. Gillian Pocock, Christopher D.Richards,"The Human Body An introduction for Biomedical and Health Sciences", Oxford University Press, USA, 2009.

1902CS353	C++ AND DATA STRUCTURES LABORATORY	L	T	P	C
		0	0	2	1

Course Objectives:

1. Learn C++ programming language.
2. Be exposed to the different data structures
3. Be familiar with applications using different data structures
4. Learn to implement stack application
5. To learn about abstract data type

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	3	-	-	-	3	2	2	3
CO2	3	3	3	-	3	-	-	-	3	2	2	3
CO3	3	3	3	-	3	-	-	-	-	-	-	3
CO4	3	3	-	-	3	-	-	-	-	-	-	3
CO5	3	-	-	-	3	-	-	-	-	-	-	3
AVG	3	3	3	-	3	-	-	-	3	2	2	3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1			
CO2			
CO3			
CO4			
CO5	-	-	2
AVG			

List of Experiments:

1. Basic Programs for C++Concepts
2. Array implementation of List Abstract Data Type(ADT)
3. Linked list implementation of List ADT
4. Cursor implementation of List ADT
5. Stack ADT-Array and linked list implementations
6. Then ext two exercises are to be done by implementing the following source files
   i. Program source files for Stack Application1
   ii. Array implementation of Stack ADT
   iii. Linked list implementation of Stack ADT
   iv. Program source files for Stack Application 2
   v. An appropriate header file for the Stack ADT should be included in (i) and (iv)
7. Implement any Stack Application using array implementation of Stack ADT(by implementing files(i) and(ii) given above) and then using linked list
8. Implementation of Stack ADT(by using files (i) and implementing file (iii))
9. Implement another Stack Application using array and linked list implementations of Stack ADT(by implementing files (iv) and using file (ii), and then by using files (iii) and (iv)
10. Queue ADT–Array and linked list implementations

	Total:	45 Hours

Additional Experiments: 1. Hash table implementation 2. Graph traversals
Course Outcomes: **After completion of the course, Student will be able to**
CO1:Identify the model of Abstract Data Type.
CO2:Calculation of algorithm efficiency and designing of recursive algorithms.
CO3:Recognize the usage of Non-Linear Data structures such as Binary Search tree, AVL search tree and Heap tree in applications.
CO4:To implement ADT for any stack application
CO5:To learn about queue ADT
Text Book:
1. Ellis Horowitz, Sartaj Sahni and Dinesh Mehta,―Fundamentals of Data Structures in C++,Galgotia Publications, 2007.
References:
1. F.Richard Gilberg, A.Behrouz. Forouzan, Data Structures, AP seudo code Approach with C.Thomson,2007.
2. M.A.Weiss, Data Structures and Algorithm Analysis in C++,Pearson Education,2009.

1902BM352	BIOCHEMISTRY AND HUMAN PHYSIOLOGY LABORATORY	L	T	P	C
		0	0	2	1

Course Objectives:

1. Estimation and quantification of biomolecules.
2. Separation of macromolecules.
3. Interpreting metabolic changes in pathological conditions.
4. Study of hemoglobin.
5. Study of anatomy using software.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	-	-	-	-	3	3	3	2
CO2	2	2	2	2	-	-	-	-	2	2	2	2
CO3	2	2	2	2	-	-	-	-	3	3	2	2
CO4	3	3	2	2	-	-	-	-	3	3	3	2
CO5	3	3	3	2	-	-	-	-	3	3	3	2
AVG	2.6	2.6	2.4	2.2	-	-	-	-	2.8	2.8	2.6	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	3	-	-
CO2	2	-	-
CO3	3	-	-
CO4	3	-	-
CO5	3	-	-
AVG	2.8	-	-

List of Experiments:

1. Study of human anatomy using A.D.A.M interactive online software.
2. Absorption spectrum of hemoglobin.
3. Bleeding time and clotting time tests.
4. Preparation of serum and plasma from blood samples.
5. Estimation of ESR, PCV, MCH, MCV, total RBC count, and hemoglobin.
6. Estimation of creatinine.
7. Estimation of urea.
8. Estimation of cholesterol.
9. Separation of amino acids by thin-layer chromatography (TLC).
10. Separation of DNA by agarose gel electrophoresis.

Total	**45 Hours**

Additional Experiments:

1. Measurement of pH of solutions using a pH meter.
2. Weber's and Rinne's tests for auditory conduction.
3. Ishihara chart test for color blindness.
4. Snellen chart test for myopia and hyperopia by letter reading.
5. Use of ophthalmoscope to view the retina.

Course Outcomes:
After completion of the course, Student will be able to

CO1:Use basic laboratory skills and apparatus to obtain reproducible data from biochemical experiments.

CO2:Separate and analyze the importance of macromolecules.

CO3:Estimate the ESR, PCV,MCH MCV total count of RBC S knowledge and Hemoglobin estimation

CO4:Estimate the levels of cholesterol, urea and creatinine.

CO5:Separation of amino acids and DNA using chromatography and electrophoresis methods.

TextBook:

1. Keith Wilson and John Walker, ―Practical Biochemistry – Principles & Techniques ― Oxford University press, 7th Edition, 2010.

References:

1. Pamela. C. Champe andRichard. A. Harvey ―Biochemistry Lippincott‟s Illustrated Reviews‖. Lippincott - Raven publishers, 6th Edition, 2013.
2. Arthur C.Guyton,"Text book of Medical Physiology",Elsevier Saunders,11th Edition,2006

S.NO	Name of the Equipment	Quantity Available (A)
1	COLORIMETER	2
2	SPECTRO PHOTO METER	1
3	PH METER	1
4	ELECTRONIC WEIGHING BALANCE	1
5	REFRIGERATOR	1
6	SDS GEL ELECTRODE	1
7	TLC PLATE 020CM	1
8	WINTROBES TUBE	4
9	CLINICAL CENTRIFUGE	1
10	MICRO SLIDES PACKETS	10
11	LANCET BOXES	10
12	MICRO SCOPE	2
13	NEUBAUR SCHAMBER	2
14	HEPARINIZED SYRINGE	1
15	HAEMOGLOBINOMETER	1
16	ELISA READER	1
17	CAPILLARY TUBE BOX	1
18	BLOOD GROUPING KIT	2
19	OPHTHALMO SCOPE	1
20	TUNING FORK 256HZ TO 512HZ	5

1902BM351	DEVICES AND CIRCUITS LABORATORY	L	T	P	C
		0	0	2	1

Course Objectives:

1. Be exposed to RL and RC circuits.
2. Become familiar with Thevenin's & Norton's theorems, Kirchhoff's Voltage Law (KVL), Kirchhoff's Current Law (KCL), and Superposition theorem.
3. Understand series and parallel resonance circuits, amplifiers, and multivibrators.
4. Observe and analyze the characteristics of diodes.
5. Design oscillators and multivibrators.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	-	-	-	-	3	3	3	2
CO2	3	2	2	2	-	-	-	-	3	3	-	2
CO3	3	3	3	2	-	-	-	-	-	3	-	2
CO4	3	3	3	3	-	-	-	-	-	-	-	-
CO5	3	3	3	3	-	-	-	-	-	-	-	-
AVG	3	2.8	2.8	2.6	-	-	-	-	3	3	3	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	2	3	-
CO2	2	3	-
CO3	2	-	-
CO4	2	-	-
CO5	-	-	-
AVG	2	3	-

List of Experiments:

1. Verification of Ohm's law, Kirchhoff's laws, and Thevenin's theorem.
2. Verification of Superposition theorem and Maximum power transfer theorem.
3. Study of rectifiers – Half-wave rectifier and Full-wave rectifier.
4. Forward and reverse characteristics of PN junction diode.
5. Forward and reverse characteristics of Zener diode.
6. Characteristics of CE Bipolar Junction Transistor (BJT).
7. Characteristics of CB Bipolar Junction Transistor (BJT).
8. Characteristics of Junction Field Effect Transistor (JFET) and Uni-Junction Transistor (UJT).
9. Design of RC phase shift oscillator.
10. Design of multivibrator.

Additional Experiments:

1. Design and Analysis of Differential Amplifier
2. Design of RC Oscillators and LC Oscillators

Course Outcomes:
After completion of the course, Student will be ablet o

CO1:Design RL and RC circuits and analyze the circuit.

CO2:Verify Thevenin& Norton theorem KVL & KCL, and Super Position Theorems
CO3:Understand the working, application and limitations of various diodes
CO4:Analyze different characteristics of transistor with different configurations
CO5:Design Oscillator, rectifier and multivibrator circuits and its applications.
Text Books: 1. Suresh Kumar K S, Electric Circuits and Networks, Pearson Education, ISBN: 9788131713907, 840 pages.
References:
1. Muhammad H. Rashid, ―Microelectronic Circuits: Analysis and Design‖, Cengage Learning, 6th Edition, 2013.
2. Robert L. Boylestad, - Electronic Devices and Circuit Theory ‖,11th Edition, 2015.
3. Robert B. Northrop, - Analysis and Application of Analog Electronic Circuits to Biomedical Instrumentation ‖, CRC Press, 2004.